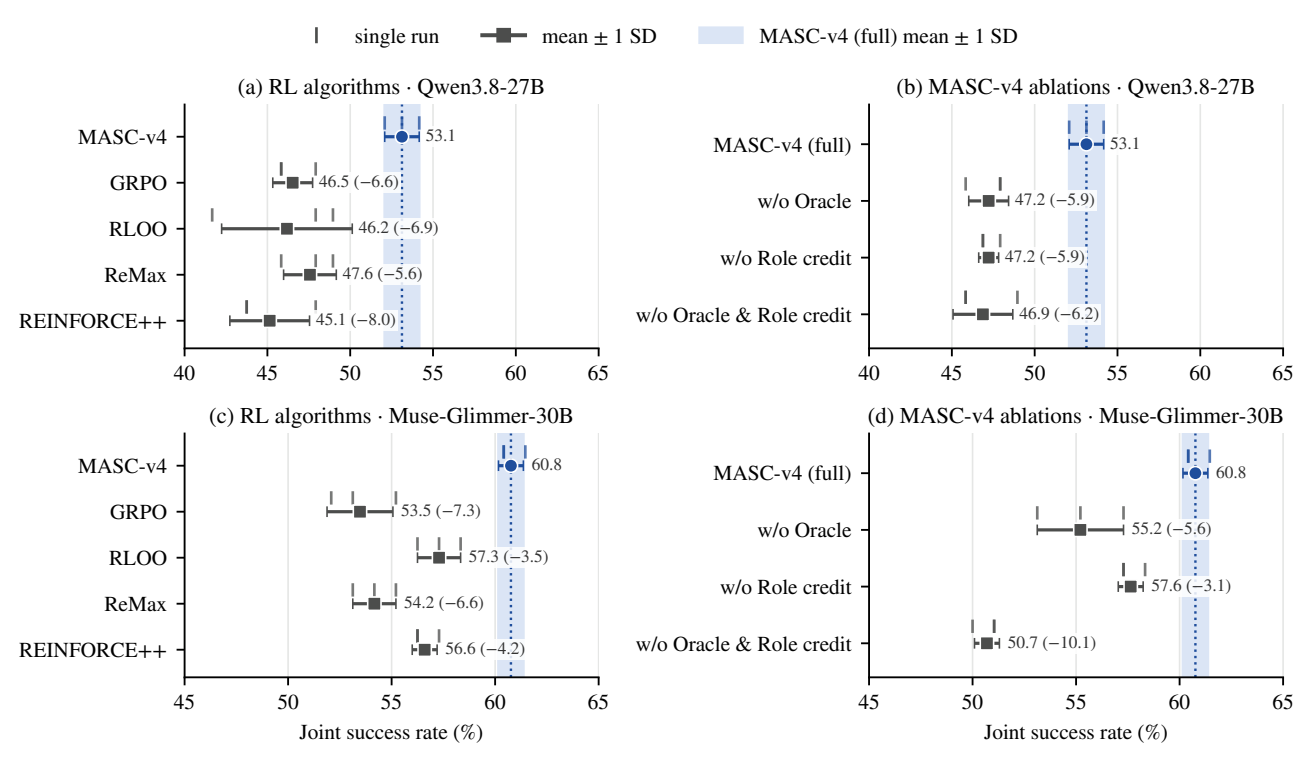
single run
mean ± 1 SD
MASC-v4 (full) mean ± 1 SD
(a) RL algorithms · Qwen3.8-27B
MASC-v4
GRPO
RLOO
ReMax
REINFORCE++
53.1
46.5 (−6.6)
46.2 (−6.9)
47.6 (−5.6)
45.1 (−8.0)
40
45
50
55
60
65
(b) MASC-v4 ablations · Qwen3.8-27B
MASC-v4 (full)
w/o Oracle
w/o Role credit
w/o Oracle & Role credit
53.1
47.2 (−5.9)
47.2 (−5.9)
46.9 (−6.2)
(c) RL algorithms · Muse-Glimmer-30B
MASC-v4
GRPO
RLOO
ReMax
REINFORCE++
60.8
53.5 (−7.3)
57.3 (−3.5)
54.2 (−6.6)
56.6 (−4.2)
45
50
55
60
65
Joint success rate (%)
(d) MASC-v4 ablations · Muse-Glimmer-30B
MASC-v4 (full)
w/o Oracle
w/o Role credit
w/o Oracle & Role credit
60.8
55.2 (−5.6)
57.6 (−3.1)
50.7 (−10.1)
Joint success rate (%)